CAIS FELLOWSHIP · AUGUST 2026

The Economics of Frontier-AI Competition

Leader profitability under open-weight catch-up

Pavel Kocourek

THE PUZZLE

OpenAI is losing money

$6B
REVENUE

$4B
INFERENCE COMPUTE

$5B
R&D TO BUILD IT

the GPT-5 “bundle” · Aug–Dec 2025

Gross margin ∼30 % — *GPT-5 never paid back its own R&D.*
Epoch AI — “Can AI companies become profitable?”, Jan 2026

THE COUNTERARGUMENT

Profitable models, unprofitable labs

- a $100M model returns $200M — profitable on its own
- the loss is the *next* model: 10× costlier, trained today
- training growth slows ⇒ profits surface

Dario Amodei — Dwarkesh Podcast · illustrative numbers

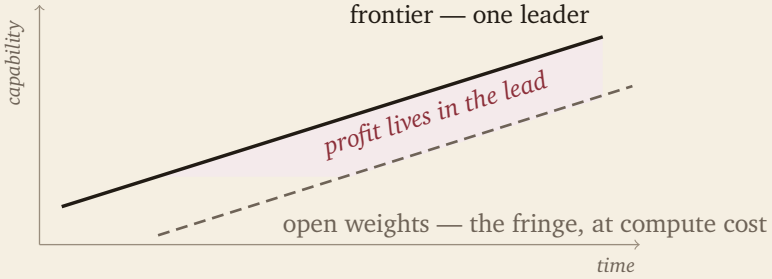
THE QUESTION

Under what conditions do frontier AI labs turn profitable?

WHY IT MATTERS

- macroeconomic stability — avoiding bubbles
- discouraging overshooting also slows firms down

The race: one leader, a following fringe



- frontier labs → modeled as one leader
- open source → a competitive fringe: the same capability, a lag later

Interactive paper

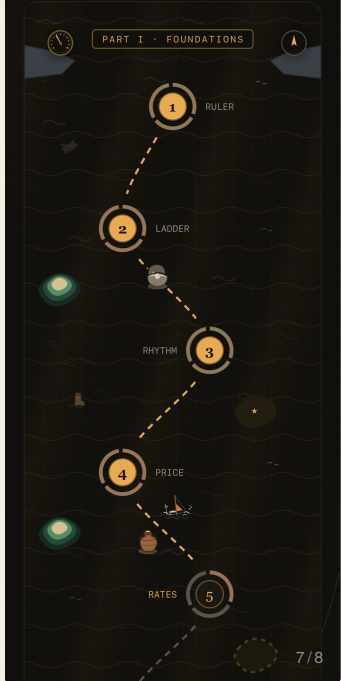
- simple economic model: extrapolate current trends, add a slowdown
- calibrate as best we can
- let the reader explore the dependence on parameter values

SIDE QUEST

Sextant

- spaced-repetition flashcards
- learn the trends in AI compute
- the weather tracks your recall

pkocourek.com/sextant/?c=ai-scale



THE REST IS

Live demo



pkocourek.com/frontier_labs_profit